Data Science for Business
ch. 6 notes: Similarity, Neighbors, and Clusters

We may want to group similar items together into clusters, for example to see whether our customer base contains groups of similar customers and what these groups have in common.

Whenever you see statements like "People who like X also like Y" or "Customers with your browsing history have also looked at ..." on Netflix, similarity is being applied.

Similarity & Distance

Firstly, **similarity and distance**. the "distance between objects" is just the mathematical way of saying "how similar are these two objects"

- We need a basic method for measuring similarity or distance. basic geometry is a good place to start:

Equation 6-1. General Euclidean distance

$$\sqrt{(d_{1,A} - d_{1,B})^2 + (d_{2,A} - d_{2,B})^2 + \ \ldots + (d_{n,A} - d_{n,B})^2}$$

- Euclidean distance is the square root of the sum of squared differences between their coordinates; it measures similarity

But this distance is just a number—it has no units, and no meaningful interpretation. It is only really useful for comparing the similarity of one pair of instances to that of another pair.

Now, **nearest neighbor reasoning**. we will use the whiskey analytics problem which is the objective of this homework to examine this.
objective (or "*objectif*" since the guys who did this came out of Montreal)**:** *how can we describe single malt scotch whiskeys as feature vectors, in such a way that we think similar whiskeys will have similar taste?*
firstly, they created some numeric features that, for any whiskey, will summarize the information in the tasting notes by defining 5 general whiskey attributes, each with many possible values

1.	**Color:** *yellow, very pale, pale, pale gold, gold, old gold, full gold, amber,* etc.	(14 values)
2.	**Nose:** *aromatic, peaty, sweet, light, fresh, dry, grassy,* etc.	(12 values)
3.	**Body:** *soft, medium, full, round, smooth, light, firm, oily.*	(8 values)
4.	**Palate:** *full, dry, sherry, big, fruity, grassy, smoky, salty,* etc.	(15 values)
5.	**Finish:** *full, dry, warm, light, smooth, clean, fruity, grassy, smoky,* etc.	(19 values)

so they found 5 similar single-malt scotches most similar by finding the closest ones.
nearest neighbors for predictive modeling: when you get a new data point, find similar past examples (nearest neighbors), and use their known outcomes to guess the new one. we use some *combining function* (like averaging) on the neighbors' known target values to give us a prediction.

- **Classification** is predicting which category or label something belongs to (e.g., yes/no, spam/not spam, default/no default)
- **Probability estimation**: we want to estimate the nearest neighbor classification's probability
- **Regression**: note that in retrieving neighbors we *don't* use the target variable in our calculations because that's what we're trying to predict.

How many neighbors and how much influence: Odd numbers are convenient for breaking ties for majority vote classification with two-class problems. The greater k is the more the estimates are smoothed out among neighbors. you *can* do k = n, but this would just predict the average over the entire dataset. The base rate is basically the overall average or proportion of an outcome in the entire dataset.

Nearest-neighbor methods often use weighted voting or similarity moderated voting such that each neighbor's contribution is scaled by its similarity.

- in the credit card example they did the reciprocal of the squared distance (basically creates a nonlinear weighting function)
- they used weighted scoring–which reduces the decision fatigue of how many neighbors to use
- so because the contribution of each neighbor is moderated by its distance, the influence of neighbors naturally decreases the farther they are from the object/instance
    - some methods avoid committing to a k by retrieving a very large number of instances (k = n) and depend upon distance weighting to moderate the influences

Regression and class probability estimation are other types of prediction tasks

Clustering

This idea of finding natural groupings in the data may be called *unsupervised segmentation*, or more simply: *clustering*.

Supervised modeling involves discovering patterns to predict the value of a specified target variable, based on data where we know the values of the target variable. Unsupervised modeling does not focus on a target variable. Instead it looks for other sorts of regularities in a set of data.

there are **two types of clustering:**

(1) hierarchical clustering: focuses on the similarities between the individual instances and how similarities link them together.

(2) clustering around centroids: A different way of thinking about clustering data is to focus on the clusters themselves—the groups of instances.

(a) The centroid is not necessarily one of the instances; it is the geometric center of a group of instances.

(b) The most popular centroid-based clustering algorithm is called k-means

(i) the "means" in "k-means" are the centroids

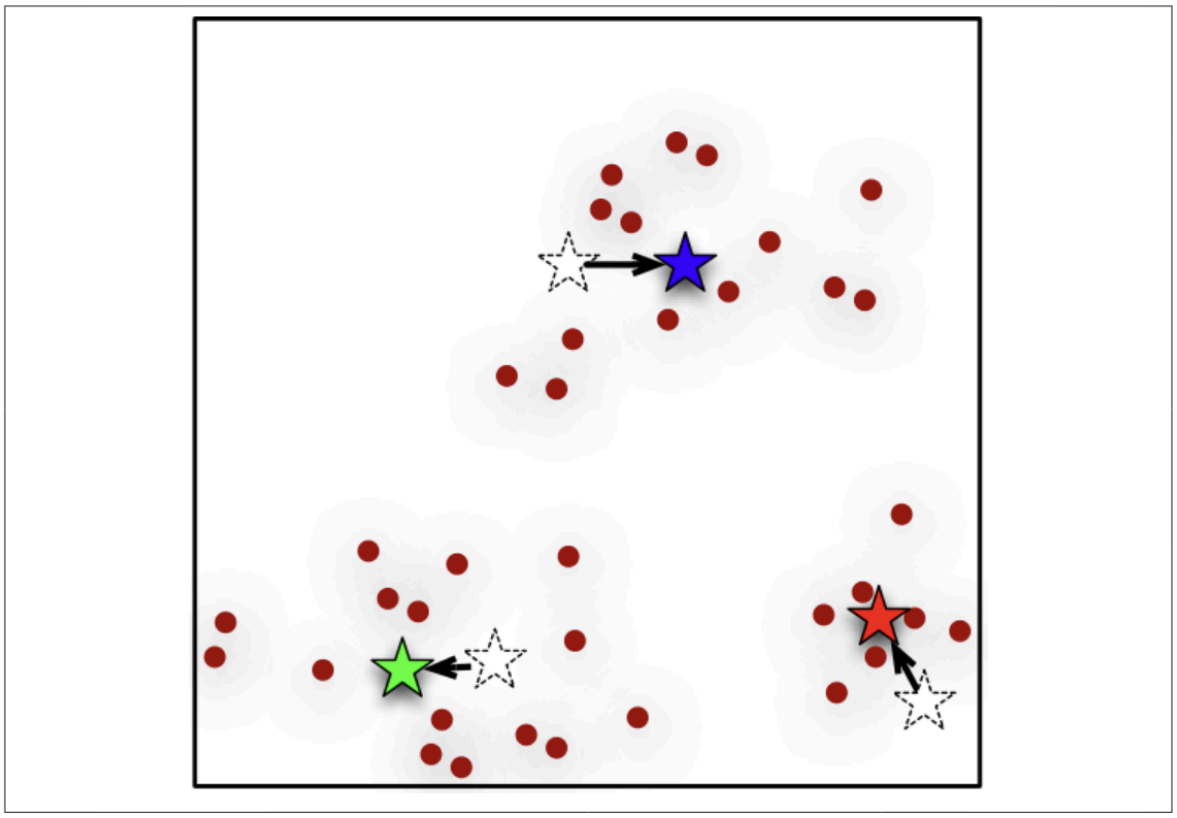


Figure 6-10. The second step of the k-means algorithm: find the actual center of the clusters found in the first step.

(ii) The *k* in k-means is simply the number of clusters that one would like to find in the data. Unlike hierarchical clustering, k-means starts with a desired number of clusters *k*.

(iii) there's an element of nearest-neighbor clustering within centroid clustering because each cluster contains those points that are nearest to its centroid (rather than to one of the other centroids)

(c) We can use preprocessing to create the cluster centroids, then the algorithm determines which data points are closest to the centroids by recalculating until there is no change in the clusters (or possibly until some other stopping criterion is met)

(d) A common concern with centroid algorithms such as k-means is how to determine a good value for k. One answer is simply to experiment with different k values and see which ones generate good results

Rule of thumb

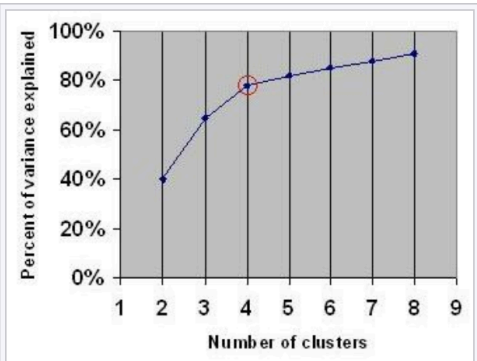
One simple rule of thumb sets the number to[1]:365

$$k \approx \sqrt{n/2}$$

with *n* as the number of objects (data points).

The Elbow Method

Another method looks at the percentage of variance explained as a function of the number of clusters: You should choose a number of clusters so that adding another cluster doesn't give much better modeling of the data. More precisely, if you graph the percentage of variance explained by the clusters against the number of clusters, the first clusters will add much information (explain a lot of variance), but at some point the marginal gain will drop, giving an angle in the graph. The number of clusters are chosen at this point, hence the "elbow criterion". This "elbow" cannot always be



Explained Variance. The "elbow" is indicated by the red circle. The number of clusters chosen should therefore be 4.

(e)

Data prep

example:

Prior to clustering, the stories underwent basic web text preprocessing, with HTML and URLs stripped out and the text case-normalized. Words that occurred rarely (fewer than two documents) or too commonly (more than 50% documents) in the corpus were eliminated, and the rest formed the *vocabulary* for the next step. Then each document was represented by a numeric feature vector using “TFIDF scores” scoring for each vocabulary word in the document. TFIDF (Term Frequency times Inverse Document